Jack Cutrara

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PROJECTS

SeatWatch | Live Reservation Booking SaaS | seat.watch 2025 - Present
TypeScript, React 19, Express, PostgreSQL, Redis, BullMQ

- Architected and shipped a production SaaS across three services (React 19 SPA, Express REST API, BullMQ worker) on Railway with PostgreSQL 17 and Redis 8, spanning ~118,000 lines of production TypeScript and ~177,000 lines of Vitest and Playwright tests.
- Built a dual-path booking engine with an automatic headless-browser fallback for graceful degradation when the primary path is unavailable.
- Engineered Redis-backed concurrency primitives: a per-user circuit breaker, nonce-guarded distributed locks with compare-and-delete release, and an atomic Lua token-bucket rate limiter, so coordination state survives restarts and deploys and no two jobs double-book the same request.
- Designed a per-user execution isolation layer with randomized 25-35 s poll scheduling, sustaining stable concurrent monitoring across users while avoiding thundering-herd load on upstream services.
- Delivered Stripe billing across four tiers with transaction-level enforcement, Clerk JWT auth, AES-256-GCM credential encryption, 20 React Email templates, and Sentry observability.

Multi-Chain EVM Trading System | Automated DeFi Trading Bot | github.com/jackc625/crypto-snipe-bot 2025 - 2026
TypeScript, Node.js, Solidity, ethers v6, SQLite, React

- Built a four-chain EVM trading runtime (Ethereum, BSC, Base, Arbitrum) with DEX-protocol differences behind discriminated unions and exhaustive compile-time dispatch checks. Shipped Ethereum-first: the other three fail closed at startup until their safety contract is deployed.
- Implemented three per-chain submission paths (Flashbots on Ethereum, bloXroute private transactions on BSC, direct EIP-1559 on Base and Arbitrum) behind a single MevStrategy interface with per-strategy health checks, over one shared nonce-safe retry and on-chain reconciliation layer replacing three divergent retry loops.
- Designed a unified exit state machine with per-token sell mutex, volatility-adjusted trailing stops, age-weighted urgency multipliers, and state persisted to SQLite every cycle. On restart it reloads every open position at its high-water mark and reconciles against the on-chain balance before resuming.

NFL Prediction System | Full-Stack ML Pipeline | github.com/jackc625/nfl-predict 2025 - Present
Python, FastAPI, XGBoost, DuckDB, scikit-learn, HTMX

- Designed a three-layer data pipeline (Bronze/Silver/Gold) ingesting play-by-play, market odds, and weather. Engineered ~195 candidate features per game (per-target selection narrows to the 20-25 each model trains on) with Pydantic hard-fail validation at ingestion and DuckDB + Parquet dual storage.
- Trained three models: logistic regression with Platt calibration for Win Probability, and XGBoost regressors for Against-the-Spread and Over/Under. All under a walk-forward temporal validation harness that hard-fails the build on detected future-data contamination.
- Built a FastAPI + Jinja2 + HTMX dashboard with Plotly charts, walk-forward backtests across 2021-2024, and parallel betting simulators (flat-stake and quarter-Kelly), automated on a weekly schedule.

PROFESSIONAL EXPERIENCE

Software Engineer, Contract | Holloway Company, Northern Virginia May 2026 - Present
JavaScript, TypeScript (Deno), React 18, TanStack Query, Base44 BaaS, Vitest

- Found and closed a critical cross-tenant data leak on a live multi-tenant platform: a test client could read 223 jobs instead of its own 1, exposing other clients' PII and financials. Re-scoped the portal from client-side filtering to server-side authorization and repaired row-level security rules failing open on empty values.
- Built the platform's test suite from 0 to 4,500+ passing checks behind a test-gated pre-commit hook and a 100%-coverage pre-push gate, pinning the highest-risk money and time calculations to the exact value with golden-master tests before extracting them into shared modules with no change to any computed value.
- Diagnosed a production regression from an over-broad bulk-archive rule the platform's tooling had silently widened, and recovered 91 wrongly-archived jobs through a backup-first, reversible, human-reviewed process, restoring the job list from 28 to 119.
- Designed an idempotent, geofenced time-clock as a pure state machine so double-taps and offline retries reconcile to exactly one payroll entry, with per-user draft isolation on shared field phones, and across-midnight or missed clock-outs flagged for human correction, never inferred.

Project Management Intern | Balfour Beatty, Chantilly, VA May 2023 - Aug 2023

- Tracked deliverables and subcontractor timelines across active construction workstreams; coordinated with engineers, subcontractors, and clients to keep multi-phase milestones on schedule.
- Led stakeholder meetings and prepared status reports for project leadership, surfacing risks and blockers in real time.

SKILLS

Languages: TypeScript, JavaScript, Python, Rust, Java, SQL, Solidity, HTML, CSS
Frontend: React, Next.js, Tailwind CSS, shadcn/ui, TanStack React Query, Zustand, Preact, HTMX
Backend & Data: Node.js, Deno, Express, FastAPI, Spring Boot, BullMQ, PostgreSQL, Redis, Prisma, SQLite, DuckDB, Base44 (BaaS), Zod, Pydantic, Supabase
Tools & DevOps: Git, GitHub Actions, Docker, Railway, Playwright, Vitest, pytest, Hardhat, Stripe, Clerk, Sentry
AI Dev Tools: Claude Code, OpenAI Codex, Cursor

EDUCATION

Bachelor of Science, Computer Science | Western Governors University Expected September 2026
Previously: Virginia Tech (2020-2023). Transferred to complete CS degree. Certifications: LPI Linux Essentials (2025)